

EDUCATION:**Massachusetts Institute of Technology, Cambridge, MA****August 2024 - May 2028***Bachelor of Science in Computer Science & Engineering and Mathematics**GPA: 5.0/5.0*

Coursework: Deep Learning, Machine Learning, Statistics, Probability, Algorithms, Computer Vision, Economics, Finance & Society

Activities: Probability & Random Variables Grader, HMMT Director, Society of Women Engineers, Undergraduate Math Association

SKILLS:

Languages & Tools: Python, Java, C/C++, TypeScript, Postgres/SQL, AWS, Pytorch, Scikit-learn, LangChain, HTML/CSS/JS

AI/ML Concepts: Feature Selection, Model Explainability, Data Visualization, Sentiment Analysis, Data Structures, Causal Graphs

PROFESSIONAL EXPERIENCE:**Aven Financial****Campbell, CA, USA***Software Engineer Intern**May 2026 – Aug 2026*

- Owning E2E development of customer-facing account servicing & payment workflows to improve cardholder retention
- Leveraging SQL, Python, TypeScript, and Playwright testing to analyze user behavior & ship scalable production features

Microsystems Technology Laboratories**Cambridge, MA, USA***Software Engineer Intern**Mar 2025 – May 2026*

- Integrating transformer-based LLMs with PostgreSQL for semantic search, query parsing, & automated patent summarization
- Developing full-stack system with Python backend, REST APIs, and HTML/CSS/JS frontend for scalable patent discovery
- Engineering data pipelines and APIs to process dataset of 100+ patents with relational mappings between 50 lab members

National University of Singapore (NUS) Computing: AI for Social Good Lab**Singapore***Human-AI Interaction Undergraduate Researcher**Mar 2025 – Aug 2025*

- Built multimodal AI integrating LLMs with TTS/STT, enabling real-time sentiment detection in human-bot interviews
- Extracted text and voice features, applying ML-driven causal graph analysis to reveal key factors behind mental health stigma

Houston Methodist Research Institute: Mathematics in Medicine Department**Houston, TX, USA***Machine Learning Research Intern**May 2023 – Jun 2025*

- Engineered and evaluated 18 tree-based, nonlinear, and linear predictive models (XGBoost, CatBoost, SVM, KNN) with Scikit-learn to classify nanoparticle toxicity using curated dataset of 8,000+ samples and 19 physicochemical features
- Implemented nested 10-fold cross-validation and GridSearch for hyperparameter optimization and robust model selection
- Utilized SHAP analysis for model explainability & feature selection; quantified generalizability (88.7% accuracy on new data)

The MITRE Corporation**McLean, VA, USA***Data Science Research Intern**Jun 2022 – Aug 2022*

- Applied statistical modeling, data analysis, and market research on iGEM datasets, literature, and SME interviews in SynBio & Computational Biology for DoD-sponsored SynBio Moonshot, generating data-driven insights to guide R&D strategies
- Published internal reports and presented to MITRE leadership, influencing strategic direction in synthetic biology innovation

PUBLICATIONS:

- Cave, J., **Christiono, A.**, et al. (2025). Rational Design of Safer Inorganic Nanoparticles Via Mechanistic Modeling-informed Machine Learning. *ACS Nano*, 19(23), 21538–21555. <https://doi.org/10.1021/acsnano.5c03590>
- **Christiono, A.**, et al. (2022). The Wiener Index: From Trees to Graphs with Many Cut-Edges [Conference presentation abstract]. 53rd Southeastern Int'l Conference in Combinatorics, Graph Theory, and Computing, Florida Atlantic University.

COMMUNITY AND TEACHING INITIATIVES:**MIT Department of EECS: Introduction to Machine Learning (6.390) Lab Assistant***Aug 2025 – Dec 2025*

- Develop lab questions reinforcing core ML concepts and encouraging broader consideration of applied ML in social contexts
- Hold office hours (4-6 hours/week) to guide students through debugging, model design, and analyses of ML results

NUS High School of Math and Science: Teacher for Advanced Calculus & Machine Learning*Jan 2025 – Feb 2025*

- Prepared and led calculus and ML courses for penultimate-year students as part of MIT's Global Teaching Labs in Singapore
- Designed slides, assessments, & interactive activities to promote discussion, analytical reasoning, & applied problem-solving

INTEGIRLS Houston: Advisor (2024-), Chapter Director (2022-2024), Web Director (2021-2022)*Aug 2021 – Jul 2024*

- Led 9 team members in outreach, fundraising, & graphic design to host free math contests for female & nonbinary students
- Participants doubled to 130+ globally; raised \$10,000+ from sponsors (doubled from prior year, including 5 new sponsors)

HONORS AND AWARDS:

- USA Olympiads: USACO Gold Division (2021-24), 5x AIME (2020-24), USABO Semifinalist (2023)
- Science Olympiad: 2x National Medalist, 25+ medals in computer science, cybersecurity, math, biology events (2022-24)
- National Center for Women & Information Technology: 2x National Honorable Mention & Houston Award Winner (2022-23)
- Writing & Scholarships: National Merit Scholar (2024), Coolidge Senator (2023), NYT STEM Writing Runner-Up (2022)